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Musical neurodynamics and the 'inner voice'

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In their Perspective, Harding et al. ask how humans experience music as more than a sequence of isolated sounds (Harding, E. E. et al. Musical neurodynamics. *Nat. Rev. Neurosci.* **26**, 293–307; 2025)[1]. They argue that their neural resonance theory (NRT) and synchronization largely explain this phenomenon, claiming that "stability of neural resonances leads to perception of musical structure" and repeatedly emphasize that "less stable [synchronized] states are attracted to and gravitate towards more stable states"[1]. Their article leaves the impression that fundamental principles based on dynamical systems can explain many basic aspects of music perception and performance. However, their theoretical claims remain inconclusive, motivating our complementary explanation.

Harding et al. frequently discuss synchrony and mode-locking, in which coupled neural oscillators synchronize to simple integer frequency ratios (for example, 1:1, 2:1 or 3:2), a phenomenon with musical implications[1]. They emphasize that synchrony is "more stable" at simpler frequency ratios. Yet they use the term stability loosely, often invoking an analogy to dynamical systems in which they claim "less stable states are attracted to and gravitate toward more stable states" (mis-citing Strogatz[2]), a claim framed in the context of neural or musical synchrony. However, such behaviour in multidimensional nonlinear systems is far from general, and for this reason many mathematicians often avoid referring to states as 'more stable' or 'less stable', treating stability instead as a binary property. Within a musical framework, such a claim would require that the authors both specify a precise notion of stability and clarify what in experiential or musical terms legitimately qualifies as a sense of 'gravitation' or 'attraction', tasks that are largely qualitative and difficult to formulate in a precise and unambiguous way.

According to the arguments in ref. 1, Harding et al. would deem 1:1 drumming between two drummers 'more stable' than 3:2. Nevertheless, mathematical models show that if the 3:2 state is 'stable'[2], it will generally not be "attracted to and gravitate towards"[1] the simpler 1:1 state, nor is it necessarily less pleasant. In fact, listeners often prefer complex ratios[3], and musically trained drummers might just as naturally shift into a 9:8 polyrhythm. The theories of Harding et al. are still being developed: integer ratios clearly appear in music, yet their place within a broader mathematical and theoretical account of musical perception – and how stability matters – remains unresolved. Moreover, the complexity of musical compositions and their internal progressions – and the human emotional responses they elicit – show that music must be more than integer ratios, as Harding et al. discuss in other publications.

Independent of these issues, synchrony still plays a crucial role in musical perception (Box 1). In ensemble performances, musicians align via repetitive structures, motifs and even brain-to-brain interactions[4]. For isolated listeners, many studies have examined how brain rhythms anticipate[1], predict[5] and entrain to external sound, but explanations for these phenomena remain incomplete. Harding et al. go beyond frequency ratios and introduce Hebbian learning and plasticity into NRT, to overcome some of these difficulties[1]. We propose the brain's 'inner voice' as a novel perspective that may also assist.

The inner voice refers to auditory imagery – tunes that run through one's head without external sound and that have gained recent attention in neuroscience[6,7]. (Extreme cases are 'earworms': unwanted songs or tunes that persist in the head.) Neuroimaging shows that auditory cortex activation during such imagery correlates with actual heard tonal relationships[6]. The inner voice functions as an internal imaginary musical instrument, simulating imaginary melodies and linking to sonic memories. Via the inner voice, composers hear printed music scores as full orchestras. Most people have this capacity in limited form, yet enough to lock onto external music. This suggests a mechanism for individual engagement with music: the inner voice attempts to entrain to, or imitate, external rhythms and melody, thereby enhancing synchrony. Although its underlying neural dynamics are intricate and poorly understood, experientially this machinery manifests as an inner voice that helps to organize perception[7]. This adds an interesting dimension to recent studies[1,5] that may help to strengthen NRT with further development.

There is a reply to this letter by Large, E. W. et al. *Nat. Rev. Neurosci.* https://doi.org/10.1038/s41583-026-01039-z (2026).

Box 1 | What makes synchronizing to music attractive?

During live performances, musicians feel that synchronizing with others is inherently 'attractive'. Consider the hand clapping of an audience: synchronization emerges through negative feedback as each individual slightly delays or advances their clap to align with their prediction[5] of the audience's next collective beat. Once synchrony is established, clapping feels effortless and self-sustaining, because minimal further feedback is required. Each person's own error-correcting feedback keeps them aligned to the shared beat, and small slips are easily corrected. When enough individuals apply 'error correction'[8] feedback, the clapping becomes 'attracted' towards a coherent, audience-wide 1:1 synchronized beat. This is the stability of synchrony. From fireflies flashing collectively in unison to inanimate pendulum clocks synchronizing simply by hanging on the same wall[2], the synchrony phenomenon is extraordinarily widespread. Hence, the human brain machinery underlying beat entrainment might be relatively simple. Comparable error-correction mechanisms have been identified in neurons[9]. Adults and children easily tap to strong beats, regardless of musical training, and the brain's basal ganglia responds robustly during beat perception[10].

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Competing interests

The authors declare no competing interests.